

Masked FER-2013: Augmented Dataset for Facial Expression Recognition

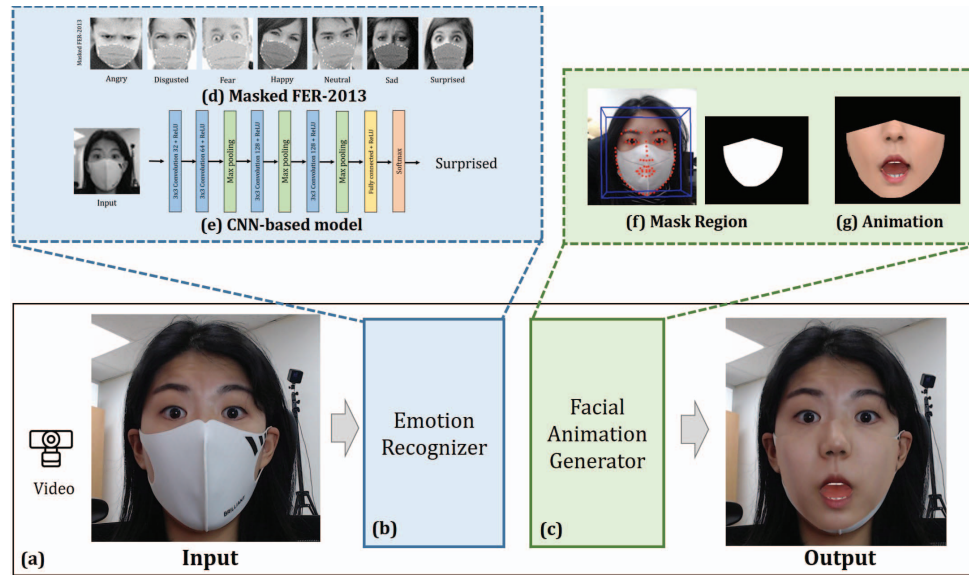
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Figure 1: Overview of the AR application use case using facial expression recognition trained with the newly synthesized dataset. (a) Diagram illustrating how our application operates in a video conference. Our application receives video streaming sources as input data and is trained with *Masked FER-2013* dataset to recognize the emotion of a masked person (b, d, e). Our application then generates an emotional facial animation and superimposes it on a real face mask (c, f, g). Thus, our application provides synthesized facial animations as output data.

ABSTRACT

This poster introduces the *Masked FER-2013* dataset that can be used to analyze facial emotions in people wearing masks and train a recognizer. The *FER-2013* dataset containing face images annotated with seven emotions was modified by synthesizing mask images into the lower portion of the face. Based on the quantitative evaluation, our dataset can improve the accuracy of *CNN*, *VGG*, and *ResNet* models' emotion recognition in masked images by a maximum of 46%. Additionally, as a use case, we present an application that recognizes the emotions of a masked person, generates emotional face animation, and augments it on a real mask during video conferencing.

Index Terms: Computing methodologies—Artificial intelligence—Computer vision—Object recognition; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

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1 INTRODUCTION

Recognizing human emotions based on facial expressions is a highly active field of research [4]. There are a lot of publicly available datasets for studying expressions on faces. Among them, the *FER-2013* is the most widely used dataset [2]. Due to the global pandemic, most people wear masks on a daily basis. According to the findings of our experiment, wearing a mask reduces the accuracy of deep learning models' emotion recognition by less than half (see Table 1 (A)). Thus, we postulated that existing systems that rely on the user's emotions for interaction might not be able to demonstrate sufficient performance for people wearing masks.

In this poster, we present the *Masked FER-2013* dataset for training facial emotion recognition on mask-wearing people. To better accommodate the current situation, we modified the *FER-2013* dataset [2] by synthesizing the face mask into the lower portion of the face. The results of our evaluation indicate that our dataset improved the accuracy of the *CNN*, *VGG19* [5], and *ResNet50* [6] model's emotion recognition by about 26% on average (see Table 1 (B)). Our dataset are available at <https://github.com/hanbin07/MaskEmotionRecognition>.

Furthermore, as a use case of our dataset, we developed an augmented reality (AR) application designed to recognize and utilize the emotions of a person wearing a mask during a video conference (see Figure 1). Our approach is to recognize the emotions of a video conference participant and augment the mask-obscured portion of the face with expressive facial animation in real-time. The following details our dataset, the use case application, and future works.

2 MASKED FER-2013

We modified the *FER-2013* dataset [2] in order to train the task of emotion recognition in mask-wearing scenarios. The original *FER-2013* dataset includes 28,709 training samples, while the public test set has 3,589 samples with seven categories: Angry, Disgusted, Fear, Happy, Neutral, Sad, and Surprised.

Creation: To synthesize an actual mask in a natural manner on the face's mask region (i.e., as opposed to simply covering the lower face), we used an open-source tool [1]. This tool can extract face landmarks, determine the mask region from the face image, and then synthesize the cropped-mask image onto the source image. However, the original *FER-2013* dataset has a small resolution of 48×48 pixels and is grayscale. Therefore, the aforementioned process of synthesizing did not produce satisfactory results. Thus we modified the procedure to meet the data specifications (e.g., size/color change) and manually excluded the failed images. As a result, we obtained 24,950 masked face images for the *Masked FER-2013* dataset (19,990 for training and 4,960 for testing). Figure 2 shows an example of our dataset.

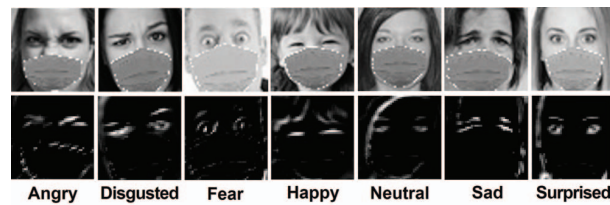


Figure 2: Examples of our *Masked FER-2013*. The first row is masked images and the second row is feature maps with seven emotion labels.

Evaluation: We developed a CNN-based model to evaluate our new dataset. The CNN model's architecture is shown in Figure 1 (e). In addition, we also have tested the performances of the conventional architectures such as *VGG19* [5] and *ResNet50* [6] trained with the *Masked FER-2013* dataset. Table 1 summarizes the comparative performance results and shows their respective accuracies increased by 13%, 20%, and 46% compared using the original *FER-2013* dataset. The confusion matrices in Figure 3 also illustrate that the recognition of the "Angry" and "Neutral" categories were particularly improved.

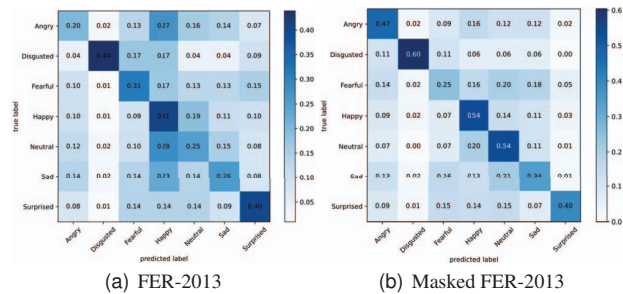


Figure 3: Confusion matrices of CNN models evaluated as test samples from our dataset (i.e., masked-face image). (a) is trained with *FER-2013*, and (b) is trained with *Masked FER-2013*.

3 USE CASE

As one use case for the *Maksed FER-2013* dataset, we developed an AR application (see Figure 1). This application is designed to resolve the difficulty in recognizing facial emotions during video conferences caused by the use of a mask. Our approach is to recognize the emotions of mask-wearing participants, generate face animations with estimated emotions, and superimpose these over the actual masks in real-time (see Figure 1). Our application consists of two parts: emotion recognizer and facial animation generator.

Emotion Recognizer: To recognize emotions of people wearing masks, our application leverages the previously mentioned CNN model trained with our dataset (see Figure 1 (d, e)). This model is lightweight for real-time usage (i.e., video conferencing) and derives emotion labels in around 20ms.

Facial Animation Generator: This part includes a series of processes for generating emotional face animations and superimposing them onto the face mask (see Figure 1 (c)). Firstly, our application analyzes the video streaming source to detect the face itself and then segments the mask region where the face animation will be augmented (see Figure 1 (f)). Next, the lower part of the face animation is generated based on the recognized emotions (see Figure 1 (g)). We also combined generated face animation with lip-sync animation (separately generated by voice input); thereby, the speaker's words can be presented visually. Finally, this facial animation is augmented in real-time while aligning with the previously detected mask area. Further details are described in our previous work [3].

This application may aid users in understanding the emotions of participants wearing masks. Nonetheless, this application serves only as an example of how to utilize the dataset. We will conduct a user study to demonstrate the system's effectiveness in future work.

4 CONCLUSION

We present a new dataset (*Masked FER-2013*), which properly overlays cropped-mask images on bare facial expression images. Our dataset can be used to recognize the emotions of a person wearing a mask. According to the quantitative results, using our dataset leads to a maximum 46% increase in the accuracy with which masked images' emotions are identified. For future work, we plan to conduct user studies and develop applications by utilizing datasets. For example, mobile AR applications can assist people with hearing disorders and children with autism spectrum disorders in interpreting the emotion of others wearing masks in face-to-face situations.

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Table 1: Summary of the accuracy results of emotion recognition for different models trained with *FER-2013* and *Masked FER-2013*.

	(A) <i>FER-2013</i>	(B) <i>Masked FER-2013</i>
CNN	30 %	43%
VGG 19 [5]	42.5 %	62.42 %
ResNet 50 [6]	18.0 %	64.94 %